Distance Vector and Link State Routing

- router inform its neighbors of topology changes periodically.
- old ARPANET routing algorithm (or known as Bellman-Ford algorithm).
- Each router maintains a Distance Vector table containing the distance between itself and ALL possible destination nodes.

- Information kept by DV router -
  - Each router has an ID
  - Associated with each link connected to a router,
  - there is a link cost (static or dynamic).
  - Intermediate hops

- Distance Vector Table Initialization -
  - Distance to itself = 0
  - Distance to ALL other routers = infinity number.

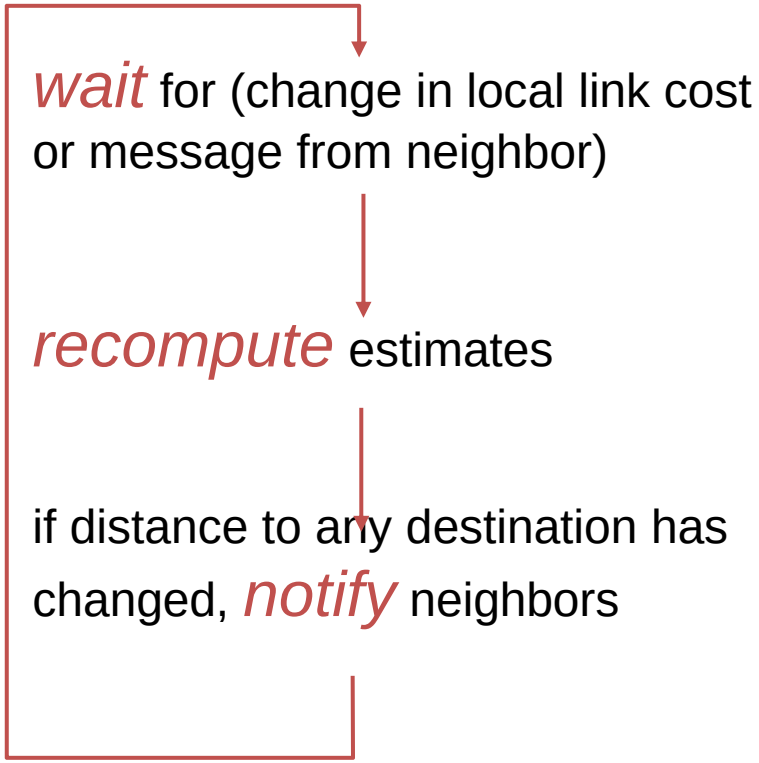
- Define distances at each node x
  - $d_x(y)$ = cost of least-cost path from x to y
- Update distances based on neighbors
  - $d_x(y) = \min \{c(x,v) + d_v(y)\}$ over all neighbors v

Iterative, asynchronous: each local iteration caused by:

- Local link cost change
- Distance vector update message from neighbor

Distributed:

- Each node notifies neighbors *only* when its DV changes
- Neighbors then notify their neighbors if necessary


wait for (change in local link cost or message from neighbor)
recompute estimates
if distance to any destination has changed, *notify* neighbors

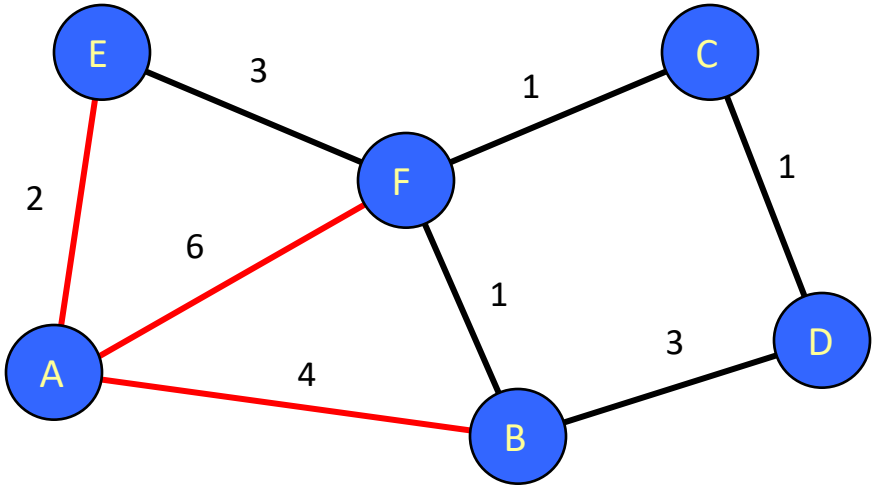

E
C
F
A
B
D
3
1
2
6
1
1
3
4

Table for A		
Dst	Cst	Hop
A	0	A
B	4	B
C	∞	–
D	∞	–
E	2	E
F	6	F

Table for B		
Dst	Cst	Hop
A	4	A
B	0	B
C	∞	–
D	3	D
E	∞	–
F	1	F

Table for C		
Dst	Cst	Hop
A	∞	–
B	∞	–
C	0	C
D	1	D
E	∞	–
F	1	F

Table for D		
Dst	Cst	Hop
A	∞	–
B	3	B
C	1	C
D	0	D
E	∞	–
F	∞	–

Table for E		
Dst	Cst	Hop
A	2	A
B	∞	–
C	∞	–
D	∞	–
E	0	E
F	3	F

Table for F		
Dst	Cst	Hop
A	6	A
B	1	B
C	1	C
D	∞	–
E	3	E
F	0	F

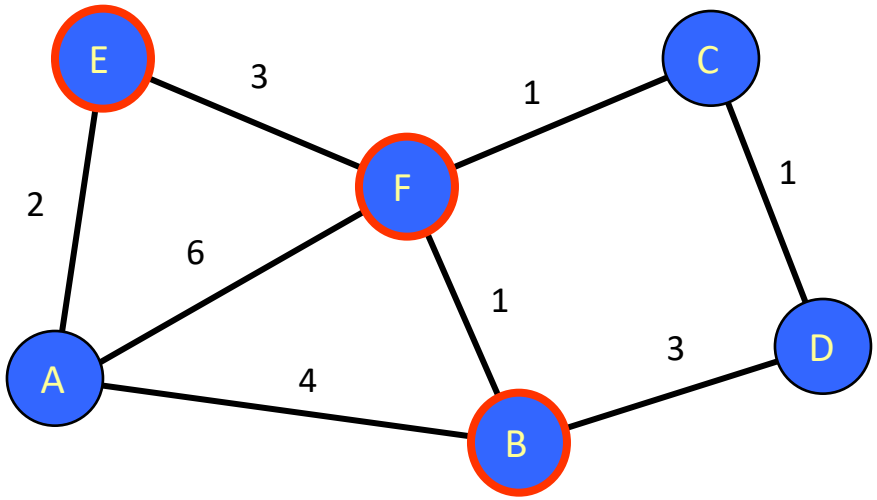

E
C
F
A
B
D
3
1
2
1
6
1
3
4

Table for A		
Dst	Cst	Hop
A	0	A
B	4	B
C	7	F
D	7	B
E	2	E
F	5	E

Table for B		
Dst	Cst	Hop
A	4	A
B	0	B
C	2	F
D	3	D
E	4	F
F	1	F

Table for C		
Dst	Cst	Hop
A	7	F
B	2	F
C	0	C
D	1	D
E	4	F
F	1	F

Table for D		
Dst	Cst	Hop
A	7	B
B	3	B
C	1	C
D	0	D
E	∞	–
F	2	C

Table for E		
Dst	Cst	Hop
A	2	A
B	4	F
C	4	F
D	∞	–
E	0	E
F	3	F

Table for F		
Dst	Cst	Hop
A	5	B
B	1	B
C	1	C
D	2	C
E	3	E
F	0	F

Final table

Table for A		
Dst	Cst	Hop
A	0	A
B	4	B
C	6	E
D	7	B
E	2	E
F	5	E

Table for B		
Dst	Cst	Hop
A	4	A
B	0	B
C	2	F
D	3	D
E	4	F
F	1	F

Table for C		
Dst	Cst	Hop
A	6	F
B	2	F
C	0	C
D	1	D
E	4	F
F	1	F

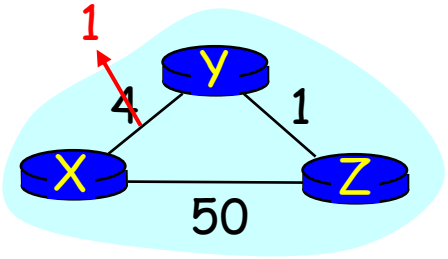
Table for D		
Dst	Cst	Hop
A	7	B
B	3	B
C	1	C
D	0	D
E	5	C
F	2	C

Table for E		
Dst	Cst	Hop
A	2	A
B	4	F
C	4	F
D	5	F
E	0	E
F	3	F

Table for F		
Dst	Cst	Hop
A	5	B
B	1	B
C	1	C
D	2	C
E	3	E
F	0	F

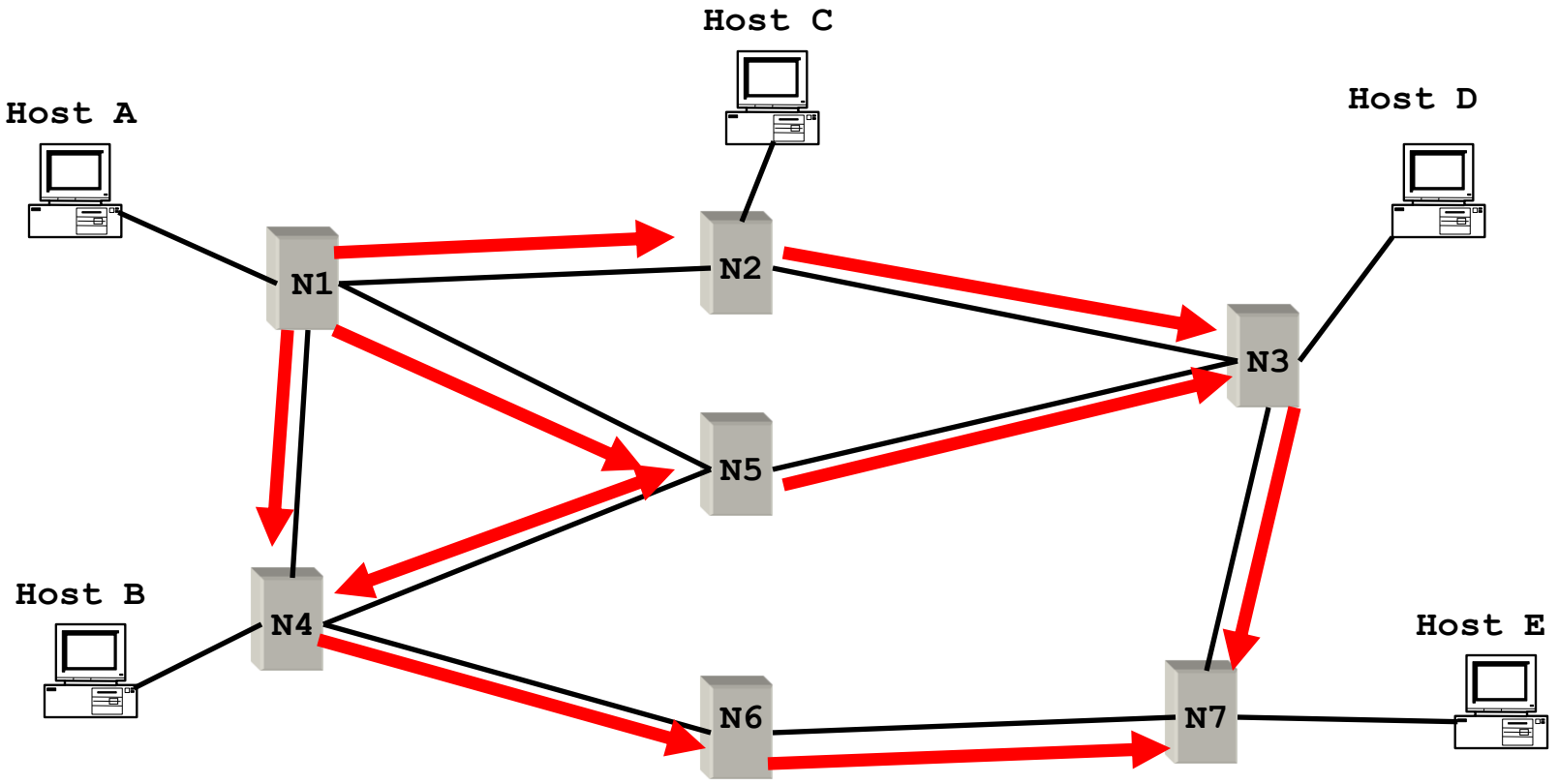
Link cost changes

- Link cost changes:
- Node detects local link cost change
- Updates the distance table
- If cost change in least cost path, notify neighbors

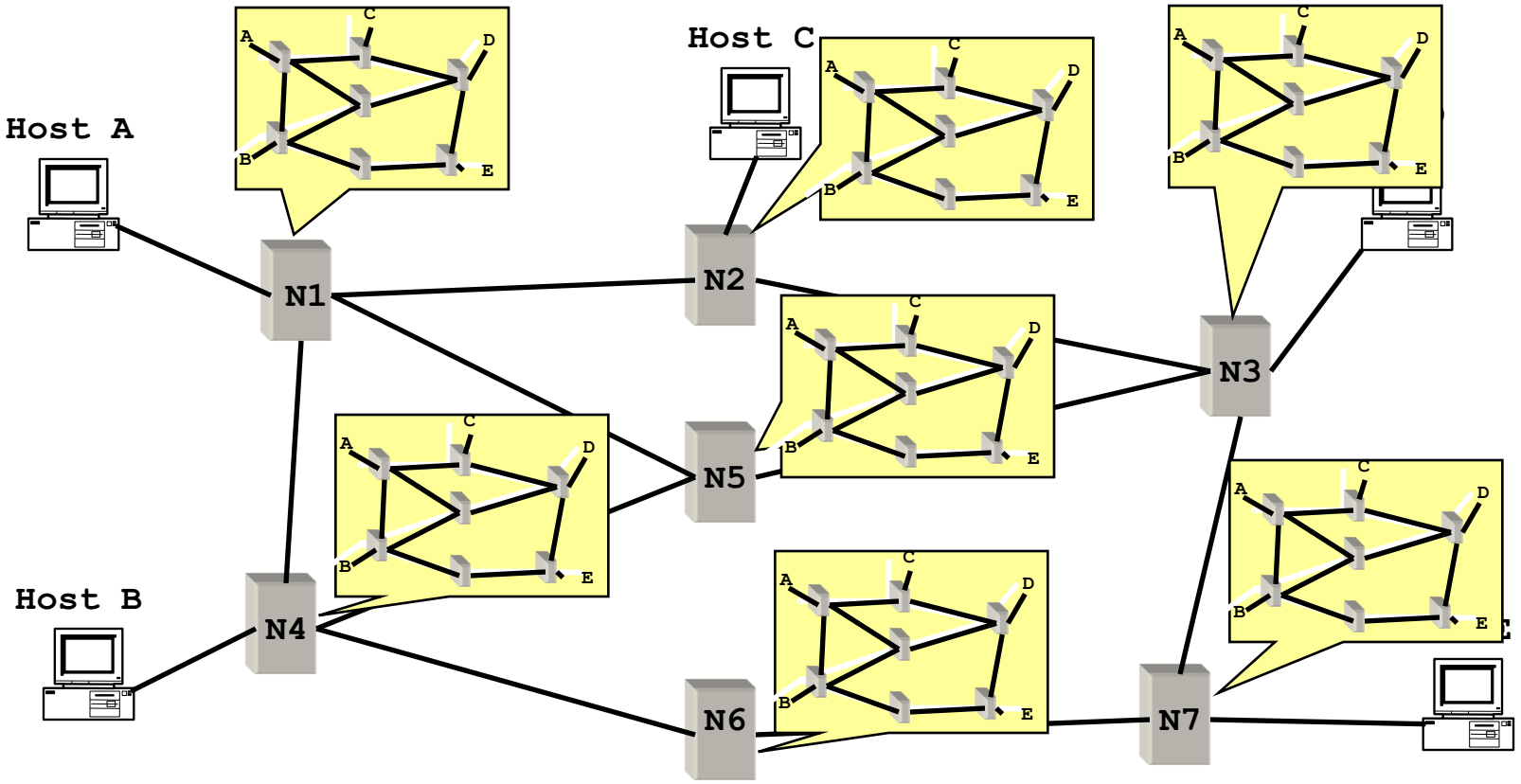


Link State: Control Traffic

- Each node floods its local information
- Each node ends up knowing the entire network topology node



Link State: Node State

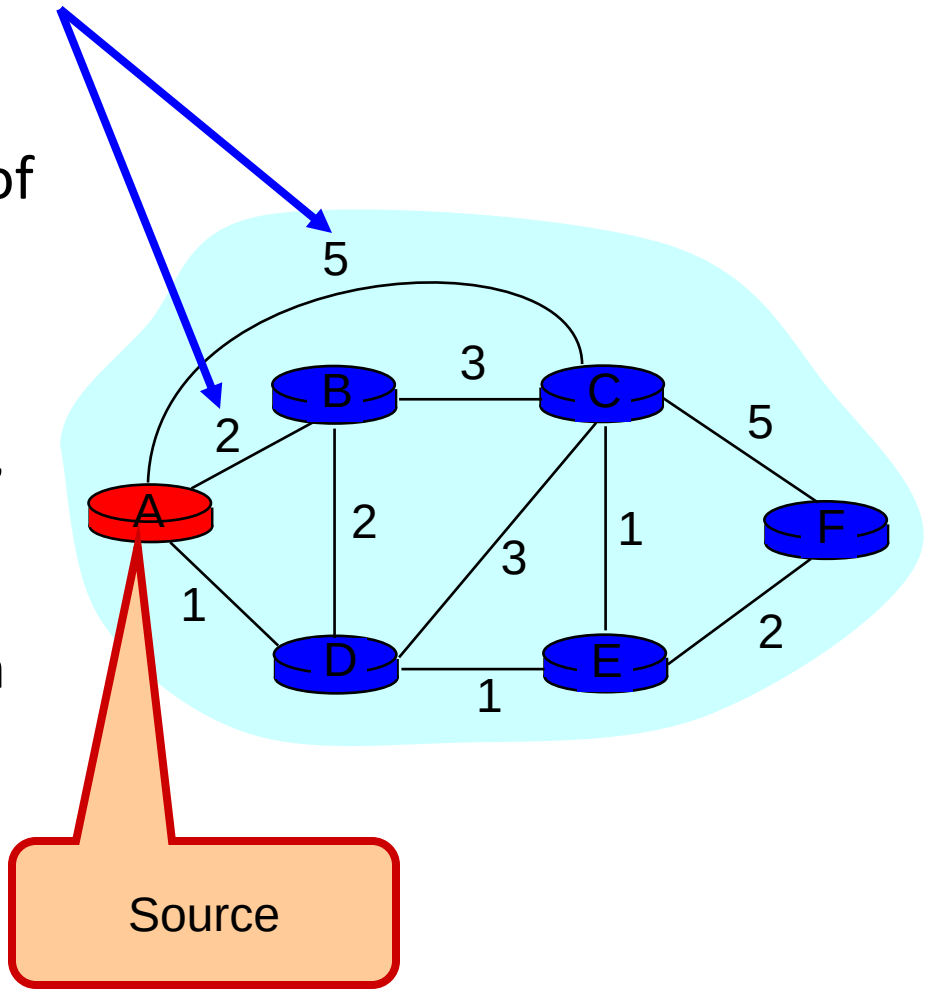


Dijkstra's Shortest Path Algorithm

- INPUT:
  - Network topology (graph), with link costs
- OUTPUT:
  - Least cost paths from one node to all other nodes
  - Produces "tree" of routes (why?)

Notation

- c(i,j): link cost from node *i* to *j*; cost infinite if not direct neighbors; **≥ 0**
- D(v): current value of cost of path from source to destination *v*
- p(v): predecessor node along path from source to *v*, that is next to *v*
- S: set of nodes whose least cost path definitively known



Dijsktra's Algorithm

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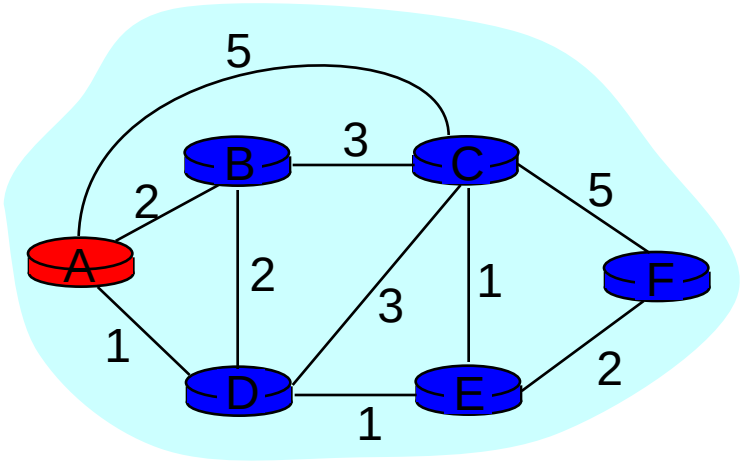
- D(v): current cost source → *v*
- p(v): predecessor node along path from source to *v*, that is next to *v*
- S: set of nodes whose least cost path definitively known

```
1  Initialization:
2    S = {A};
3    for all nodes v
4      if v adjacent to A
5        then D(v) = c(A,v);
6        else D(v) = ∞;
7
8  Loop
9      find w not in S such that D(w) is a minimum;
10     add w to S;
11     update D(v) for all v adjacent to w and not in S:
12       if  D(w) + c(w,v) < D(v) then
             // w gives us a shorter path to v than we've found so far
13           D(v) = D(w) + c(w,v); p(v) = w;
14 until all nodes in S;
```

Example: Dijkstra's Algorithm

Step	start S	D(B),p(B)	D(C),p(C)	D(D),p(D)	D(E),p(E)	D(F),p(F)
0	A	2,A	5,A	1,A	∞	∞
1						
2						
3						
4						
5						

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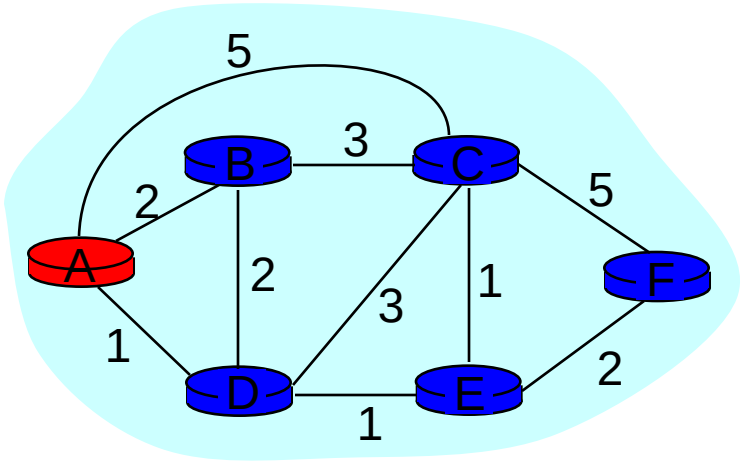
1						
2						
3						
4						
5						

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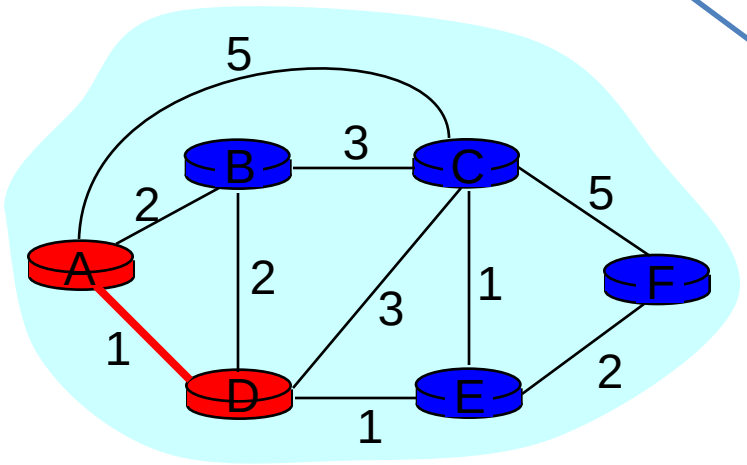
0	A	2,A	5,A	1,A	∞	∞
1	AD					
2						
3						
4						
5						

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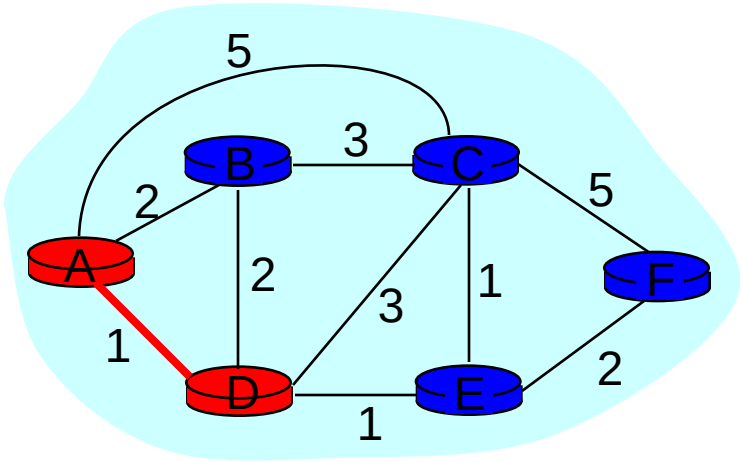
0	A	2,A	5,A	1,A	∞	∞
1	AD		4,D		2,D	
2						
3						
4						
5						

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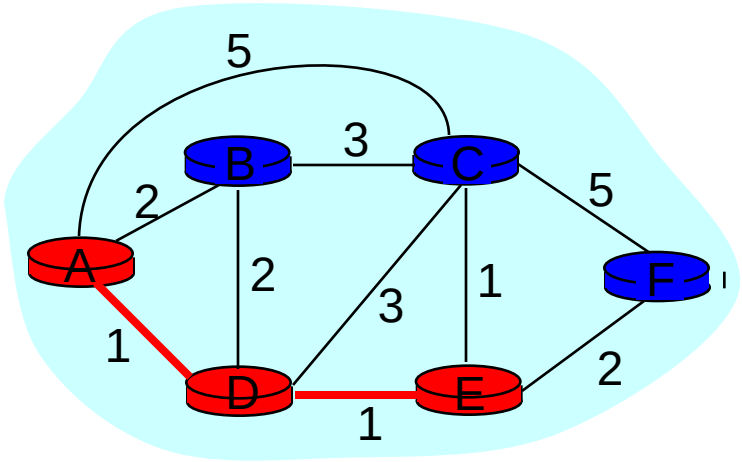
2	ADE		3,E			4,E
3						
4						
5						

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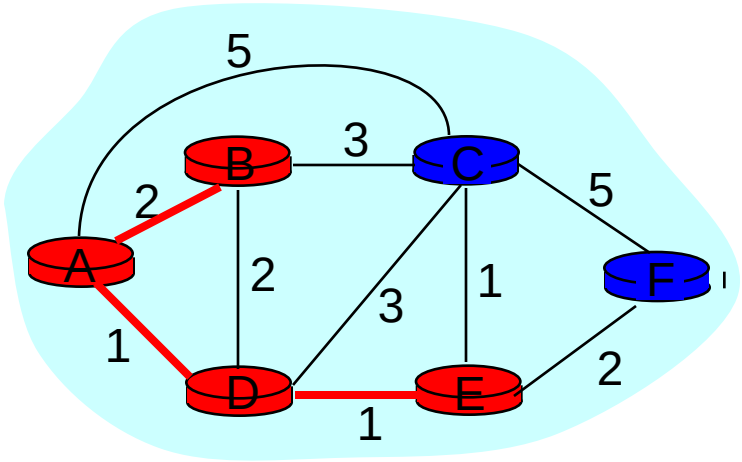
1	AD		4,D		2,D	
2	ADE		3,E			4,E
3	ADEB					
4						
5						

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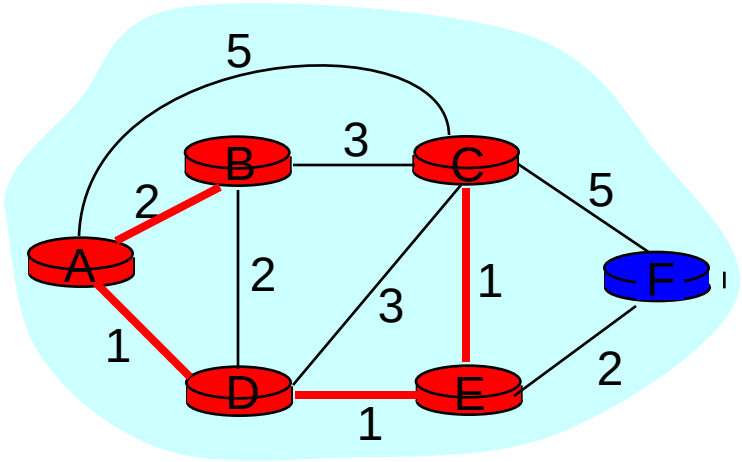
0	A	2,A	5,A	1,A	∞	∞
1	AD		4,D		2,D	
2	ADE		3,E			4,E
3	ADEB					
4	ADEBC					
5						

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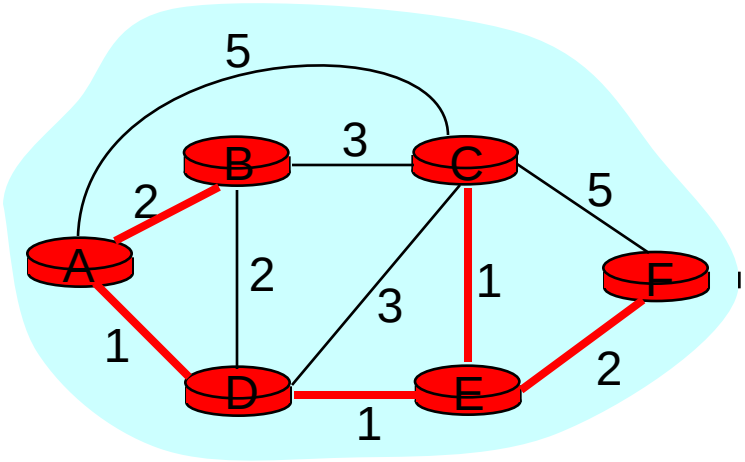
0	A	2,A	5,A	1,A	∞	∞
1	AD		4,D		2,D	
2	ADE		3,E			4,E
3	ADEB					
4	ADEBC					
5	ADEBCF					

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Example: Dijkstra's Algorithm

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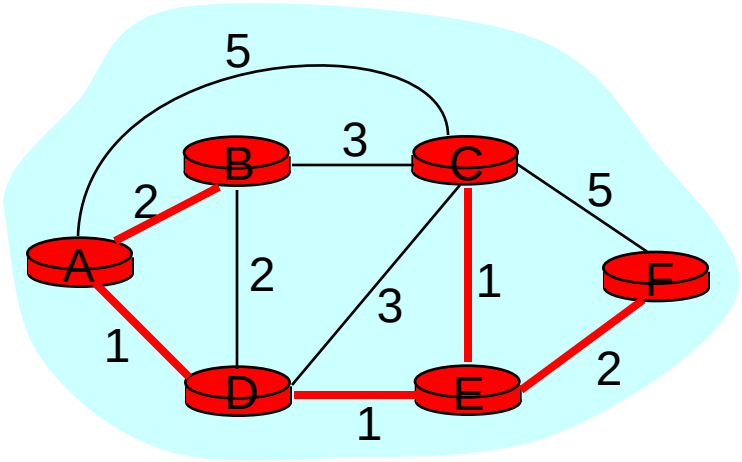
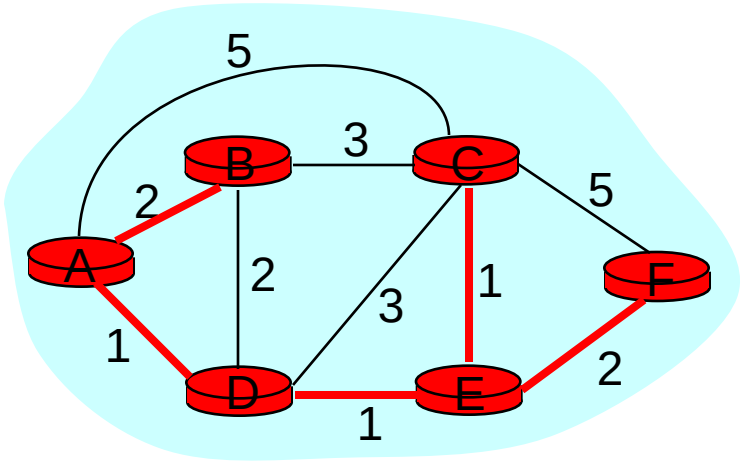
0	A	2,A	5,A	1,A	∞	∞
1	AD		4,D		2,D	
2	ADE		3,E			4,E
3	ADEB					
4	ADEBC					
5	ADEBCF					



To determine path A → C (say), work backward from C via p(v)

The Forwarding Table

- Running Dijkstra at node A gives the shortest path from A to all destinations
- We then construct the *forwarding table*



Destination	Link
B	(A,B)
C	(A,D)
D	(A,D)
E	(A,D)
F	(A,D)

Complexity

- How much processing does running the Dijkstra algorithm take?
- Assume a network consisting of N nodes
  - Each iteration: check all nodes w not in S
  - N(N+1)/2 comparisons: $O(N^2)$
  - More efficient implementations: O(N log(N))